

U.G.C. NET EX COMPUTER (Solved with Expl

1. How many different equivalence relations with exactly three different equivalence classes are there on a set with five elements?

- (a) 10 (b) 15
(c) 25 (d) 30

Ans : (c) The set of equivalence relations on a set are in direct bijection with the set of partitions on a set. We know that the only partition types for a 5 element set are of the form $(1 + 1 + 3)$ or $(1 + 2 + 2)$ of the first type, pick the three in the triple, ${}^5C_3 = 10$ in case of the second type, first pick which is the singleton and then pick which is paired with smallest remaining, i.e., $5 \times {}^3C_1 = 15$ of this for a total of $10 + 15 = 25$.

2. The number of different spanning trees in complete graph, K_4 and bipartite graph, $K_{2,2}$ have _____ and _____ respectively.

- (a) 14, 14 (b) 16, 14
(c) 16, 4 (d) 14, 4

Ans : (c) The number of spanning trees in complete graph with n vertices.

$$= n^{n-2} = 4^{4-2} = 16$$

A complete bipartite graph $k_{m,n}$ has $m^{n-1} n^{m-1}$ spanning trees $= 2^1 \cdot 2^1 = 4$.

3. Suppose that R_1 and R_2 are reflexive relations on a set A.

Which of the following statements is correct?

- (a) $R_1 \cap R_2$ is reflexive and $R_1 \cup R_2$ is irreflexive.
(b) $R_1 \cap R_2$ is irreflexive and $R_1 \cup R_2$ is reflexive.
(c) Both $R_1 \cap R_2$ and $R_1 \cup R_2$ are reflexive.
(d) Both $R_1 \cap R_2$ and $R_1 \cup R_2$ are irreflexive.

Ans : (c) If R_1 and R_2 are reflexive relations on a set A. $R_1 \cup R_2$ is reflexive because.

$$\Delta \subseteq R_1 \subseteq R_1 \cup R_2 \text{ (or } \Delta \subseteq R_2 \subseteq R_1 \cup R_2)$$

$R_1 \cap R_2$ is also reflexive, because

$$\Delta \subseteq R_1 \wedge \Delta \subseteq R_2 \Rightarrow \Delta \subseteq R_1 \cap R_2$$

4. There are three cards in a box. Both sides of one card are black, both sides of one card are red, and the third card has one black side and one red side. We pick a card at random and observe only one side.

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What is the probability that the opposite side is the same colour as the one side we observed?

- (a) $3/4$ (b) $2/3$
 (c) $1/2$ (d) $1/3$

Ans : (b) There are three cards, such that the sides of cards are:

$\{\{\text{Black, Black}\}, \{\text{Red, Red}\}, \{\text{Black, Red}\}\}$

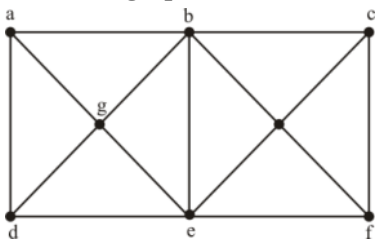
There are total 3 sample space as given above.

Now favorable cases are:

$\{\{\text{Black, Black}\}, \{\text{Red, Red}\}\}$ only 2 cases.

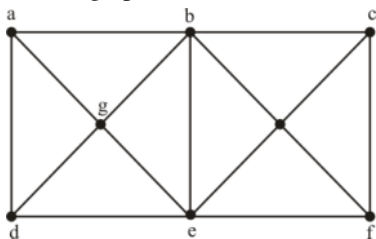
Therefore, required probability is $2/3$.

5. A clique in a simple undirected graph is a complete subgraph that is not contained in any larger complete subgraph. How many cliques are there in the graph shown below?



- (a) 2 (b) 4
 (c) 5 (d) 6

Ans : (c) Given graph is



Complete subgraph means “each vertex should be connected with all other vertices in the subgraph”. Such subgraph are:

- (i) $\{b, c, e, f\}$ (ii) $\{a, d, g\}$
 (iii) $\{a, b, g\}$ (iv) $\{b, e, g\}$
 (v) $\{d, e, g\}$

There are only 5 subgraphs.

6. Which of the following logic expressions is incorrect

- (a) $1 \oplus 0 = 1$ (b) $1 \oplus 1 \oplus 1 = 1$
 (c) $1 \oplus 1 \oplus 0 = 1$ (d) $1 \oplus 1 = 0$

Ans : (c) With EX-OR operation, odd number of 1's returns 1 and even number of 1's return 0.

7. The IEEE-754 double-precision format to represent floating point number, has a length of _____ bits.

- (a) 16 (b) 32
(c) 48 (d) 64

Ans : (d) Double precision floating point format is a computer number format that occupies 8 bytes (64 bits) in computer memory and represents a wide, dynamic range of values by using a floating point.

- Sign bit (1 bit)
- Exponent (11 bit)
- Significant precision (53 bits)/52 explicitly stored.

8. Simplified Boolean equation for the following truth table is:

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

- (a) $F = y\bar{z} + \bar{y}z$ (b) $F = x\bar{y} + \bar{x}y$
(c) $F = \bar{x}z + x\bar{z}$ (d) $F = \bar{x}z + x\bar{z} + xyz$

Ans : (c) Function f will be:

$$\begin{aligned} f &= x'y'z + x'yz + xy'z' + xyz' \\ &= x'z(y' + y) + xz'(y' + y) \\ &= x'z + xz' \end{aligned}$$

9. The simplified form of a Boolean equation $(A\bar{B} + A\bar{B}C + AC)(\bar{A}\bar{C} + \bar{B})$ is:

- (a) $A\bar{B}$ (b) $A\bar{B}C$
(c) $\bar{A}B$ (d) ABC

Ans : (a) Given Boolean expression is:

$$\begin{aligned} &= (A\bar{B}' + A\bar{B}'C + AC)(A'C' + B') \\ &= (A\bar{B}'(1+C) + AC)(A'C' + B') \\ &= (A\bar{B}' + AC)(A'C' + B') \\ &= AA'B'C' + A\bar{B}' + AA'CC' + AB'C \\ &= A\bar{B}' + AB'C \\ &= A\bar{B}'(1+C) \\ &= A\bar{B}' \end{aligned}$$

10. In a positive-edge-triggered JK flip-flop, if J and K both are high then the output will be _____ on the rising edge of the clock.
- (a) No change (b) Set
(c) Reset (d) Toggle

Ans : (d) Racing is problem of toggling more than once when the inputs are $J = 1$ and $K = 1$. The proper combination of propagation delay and edge triggering prevents the racing problem.

11. Given $i = 0, j = 1, k = -1$
 $x = 0.5, y = 0.0$
 What is the output of the following expression in C language?
 $x * y < i + j \parallel k$
- (a) -1 (b) 0
(c) 1 (d) 2

Ans : (c) Given expressions are:
 $i = 0, j = 1, k = -1, x = 0.5, y = 0.0$ and
 $= x * y < i + j \parallel k$
 $= ((x * y) < (i + j)) \parallel k$
 $= (((0.5 * 0.0) < (0 + 1)) \parallel -1)$
 $= ((0 < 1) \parallel -1)$
 $= 1$

12. The following statement in ‘C’
`int (*f( ))[ ]:`
declares
- (a) a function returning a pointer to an array of integers.
 (b) a function returning an array of pointers to integers.
 (c) array of functions returning pointers to integers.
 (d) an illegal statement.

Ans : (a) `int (*f( )) [ ]`
 A function f returning a pointer to an array of integers.

13. Which one of the following is correct, when a class grants friend status to another class?
- (a) The member functions of the class generating friendship can access the members of the friend class.
 (b) All member functions of the class granted friendship have unrestricted access to the members of the class granting the friendship.
 (c) Class friendship is reciprocal to each other.
 (d) There is no such concept.

Ans : (b) The function of a class which is given friend status by another class can access all its private, protected and public members. Class friendship is not commutative nor transitive unless explicitly specified.

14. When a method in a subclass has the same name and type signatures as a method in the superclass, then the method in the subclass _____ the method in the superclass.

- (a) Overloads (b) Friendships
(c) Inherits (d) Overrides

Ans : (d) In a class hierarchy, when a method in a subclass has the name and type signature as a method in its superclass, then the method in the subclass is said to override the method in the superclass.

15. What is the value returned by the function f given below when $n = 100$?

```
int f(int n)
{ if (n == 0) then return n;
  else
    return n + f(n - 2);
}
```

- (a) 2550 (b) 2556
(c) 5220 (d) 5520

Ans : (a) The given function is a recursive function which takes starting value as 100. It keeps on adding the present value and the value obtained by calling the function $f(n - 2)$ where n is current value. It stops at $n = 0$ where it returns 0. As a result, it forms an arithmetic series, where $a = 100$ and $d = 2$ and $l = 2$. So sum of series is

$$S_{100} = \frac{n(a_1 + a_n)}{2} = \frac{50(2 + 100)}{2} = 2550$$

16. In RDBMS, the constraint that no key attribute (column) may be NULL is referred to as:

- (a) Referential integrity
(b) Multi-valued dependency
(c) Entity Integrity
(d) Functional dependency

Ans : (c) The **entity integrity constraint** states that primary keys cannot be null. There must be a proper value in the primary key field.

The basis of referential integrity is foreign keys. A foreign key in one table references a primary key in another table.

17. Which of the following statement(s) is/are FALSE in the context of Relational DBMS?

- I. Views in a database system are important because they help with access control by allowing users to see only a particular subset of the data in the database.
II. E-R diagrams are useful to logically model concepts.

III. An update anomaly is when it is not possible to store information unless some other, unrelated information is stored as well.

IV. SQL is a procedural language.

- (a) I and IV only (b) III and IV only
(c) I, II and III only (d) II, III and IV only

Ans : (b) (i) Views in a database system are important because:

- They help provide data independence.
- They allow the schema to change without forcing existing applications to be recompiled.
- They help with access control by allowing users to see only a particular subset of the data in the database.

(ii) A conceptual ER model may be used as the foundation for one or more logical data models.

(iii) That's not update anomaly but it is insertion anomalies.

(iv) SQL is not procedural language.

Only statements (iii) and (iv) are false.

18. In a relational database model, NULL values can be used for all but which one of the following?

- (a) To allow duplicate tuples in the table by filling the primary key column(s) with NULL.
- (b) To avoid confusion with actual legitimate data values like 0 (zero) for integer columns and "" (the empty string) for string columns.
- (c) To leave columns in a tuple marked as "unknown" when the actual value is unknown.
- (d) To fill a column in a tuple when that column does not really "exist" for that particular tuple.

Ans : (a) The **SQL** null is the term used to represent a missing value. A null value in a table is a value in a field that appears to be blank. A field with a null value is a field with no value.

19. Consider the following two commands C1 and C2 on the relation R from an SQL database:

C1: drop table R;

C2: delete from R;

Which of the following statements is TRUE?

I. Both C1 and C2 delete the schema for R.

II. C2 retains relation R, but deletes all tuples in R.

III. C1 deletes not only all tuples of R, but also the schema for R.

- (a) I only (b) I and II only
(c) II and III only (d) I, II and III

Ans : (c) Drop command remove all tuples of the relation as well as schema also.

Delete command in SQL removes tuples from the relation but retain the schema of table in database.

20. Consider the following database table having A, B, C and D as its four attributes and four possible candidate keys (I, II, III, IV) for this table:

A	B	C	D
a_1	b_1	c_1	d_1
a_2	b_2	c_3	d_2
a_3	b_3	c_1	d_3

I: {B}

II: {B, C}

III: {A, D}

IV: {C, D}

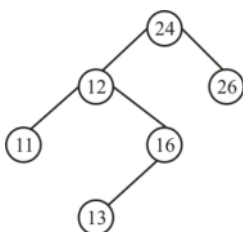
If different symbols stand for different values in the table (e.g., d_1 is definitely not equal to d_2), then which of the above could not be the candidate key for the database table?

- (a) I and III only (b) III and IV only
(c) II only (d) I only

Ans : (c) Attribute B has distinct tuples. So, B is candidate key and {BC} is superkey because a candidate key is minimal superkey.

Key {A, D} and {C, D} has distinct tuples, so, both are candidate key.

21. Consider the following binary search tree:

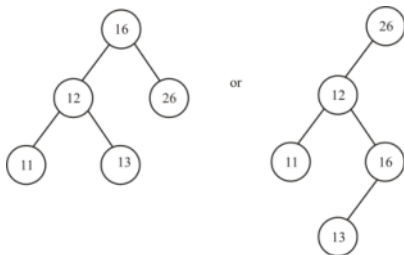


If we remove the root node, which of the node from the left subtree will be the new root?

- (a) 11 (b) 12
(c) 13 (d) 16

Ans : (d) If we delete root node of a binary search tree, then it can be replace by inorder successor or inorder predecessor of the node.

So, either 16 or 26 can be root of the tree.



Which of the following statements is correct?

25. In how many ways can the string

$$A \cap B - A \cap B - A$$

Be fully parenthesized to yield an infix expression?

- (a) 15 (b) 14
(c) 13 (d) 12

Ans : (b) Number of ways we can parenthesis n operands is catalun number.

$$= \frac{{}^{2n}C_n}{n+1}$$

Given number of total operators ($n = 4$).

$$\text{So, } = \frac{{}^8C_4}{5} = 14$$

Note: All operator works as nodes of binary search tree.

26. A multiplexer combines four 100-Kbps channels using a time slot of 2 bits. What is the bit rate?

- (a) 100 Kbps (b) 200 Kbps
(c) 400 Kbps (d) 1000 Kbps

Ans : (c) bit rate = no. of channels $\times$ Input bit rate
 $= 4 \times 100 \text{ kbps} = 400 \text{ kbps}$

27. In a fully-connected mesh network with 10 computers, total _____ number of cables are required and _____ number of ports are required for each device.

- (a) 40, 9 (b) 45, 10
(c) 45, 9 (d) 50, 10

Ans : (c) Since, each in mesh network total number of cables are required

$${}^nC_2 = {}^{10}C_2 = 45$$

Each device needs to be connected to 9 other devices.
 So, each device needs to have 9 ports.

28. In TCP/IP Reference model, the job of _____ layer is to permit hosts to inject packets into any network and travel them independently to the destination.

- (a) Physical (b) Transport
(c) Application (d) Host-to-network

Ans : (*) The job of Internet layer is to permit hosts to inject packets into any network and have them travel independently to the destination (potentially on a different network).

All option is true, as given by official keys.

29. If there are N people in the world and are using secret key encryption/decryption for privacy purpose, then number of secret keys required will be:

- (a) N (b) $(N - 1)$
(c) $\frac{N(N-1)}{2}$ (d) $\frac{N(N+1)}{2}$

Ans : (c) Each pair of users must have a secret key. If N people in world want to use this method, there needs to be ${}^nC_2 = \frac{n(n-1)}{2}$ secret keys.

30. Optical fiber uses reflection to guide light through a channel, in which angle of incidence is _____ the critical angle.
 (a) equal to (b) less than
 (c) greater than (d) less than or equal to

Ans : (c) If the **angle of incidence** is equal to the critical angle, the light beam bends along the interface. When a **beam of light travels** through media of two different densities, if the angle of incidence is greater than the critical angle, reflection occurs.

31. The number of strings of length 4 that are generated by the regular expression $(0 | \epsilon) 1^+ 2^* (3 | \epsilon)$, where $|$ is an alternation character, $\{+, *\}$ are quantification characters, and ϵ is the null string, is:
 (a) 08 (b) 10
 (c) 11 (d) 12

Ans : (d) Given, regular expression is:
 $R = (0 | \epsilon) 1^+ 2^* (3 | \epsilon)$
 Number of strings of length 4 are:
 $\{1111, 1113, 1222, 1122, 1112, 1223, 1123, 0111, 0113, 0122, 0112, 0123\}$
 Total number of string 12.

32. The content of the accumulator after the execution of the following 8085 assembly language program, is:
MVI A, 42H
MVI B, 05H
UGC: ADD B
DCR B
JNZ UGC
ADI 25H
HLT
 (a) 82 H (b) 78 H
 (c) 76 H (d) 47 H

Ans : (c) A is taking accumulator
 $A = 4 \times 16 + 2 = 66$ in decimal
 $B = 5$ in decimal
 Loops runs from $B = 5$ to 1 therefore,
 $= 66 + 5 + 4 + 3 + 2 + 1 = 81$
 Then, $81 + 25 H$
 $= 81 + 37$ (35 H in decimal) = 118
 $\Rightarrow (118)_{10} = 76 H$

33. In _____, the bodies of the two loops are merged together to form a single loop provided that they do not make any references to each other.

- (a) Loop unrolling
- (b) Strength reduction
- (c) Loop concatenation
- (d) Loop jamming

Ans : (d) (i) Loop fusion (Jamming) is a compiler optimization and loop transformation which replaces multiple loops with a single one. It is possible when two loops iterate over the same range and do not reference each other's data.

(ii) Loop unrolling (loop unwinding) is a loop transformation technique that attempts to optimize a program's execution speed at the expense of its binary size, which is an approach known as the space time tradeoff.

34. Which of the following is not typically a benefit of dynamic linking?

- I. Reduction in overall program execution time.
- II. Reduction in overall space consumption in memory.
- III. Reduction in overall space consumption on disk.
- IV. Reduction in the cost of software updates.

- (a) I and II
- (b) I only
- (c) II and III
- (d) IV only

Ans : (b) (i) A dynamic linker is the part of an operating system that loads and links the shared libraries needed by an executable when it is executed (at "run time"), by copying the context of libraries from present storage to RAM, and filling jump tables and relocating pointers.

(ii) Static linker has reduction in overall program execution time than dynamic linker.

35. Which of the following is FALSE?

- (a) The grammar $S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$, where S is the only non-terminal symbol and ϵ is the null string, is ambiguous.
- (b) SLR is powerful than LALR.
- (c) An LL(1) parser is a top-down parser.
- (d) YACC tool is an LALR(1) parser generator.

Ans : (b)

(a) True, since string ϵ has infinite many parse tree.

(b) False, since LALR(1) grammars are more powerful than also true.

(c) LL(1) parser is a top-down parser.

(d) YACC tool is an LALR(1) parser generator.

36. Consider the reference string:

0 1 2 3 0 1 4 0 1 2 3 4

If FIFO page replacement algorithm is used, then the number of page faults with three page frames and four page frames are _____ and _____ respectively.

- (a) 10, 9
- (b) 9, 9
- (c) 10, 10
- (d) 9, 10

Ans : (d) Given reference string is 0, 1, 2, 3, 0, 1, 4, 0, 1, 2, 3, 4.

(i) FIFO (first in first out) algorithm with 3 page frames.

0	1	2	3	0	1	4	0	1	2	3	4
0	0	0	3	3	3	4			4	4	
	1	1	1	0	0	0			2	2	
		2	2	2	1	1			1	3	

M M M M M M M M M M

Total number of page faults = 9

(ii) FIFO with 4 page frames.

0	1	2	3	0	1	4	0	1	2	3	4
0	0	0	0			4	4	4	4	3	3
	1	1	1			1	0	0	0	0	4
		2	2			2	2	1	1	1	1
			3			3	3	3	2	2	2

M M M M M M M M M M

Total number of page faults = 10.

37. Suppose there are four processes in execution with 12 instances of a Resource R in a system. The maximum need of each process and current allocation are given below:

Process	Max. Need	Current Allocation
P ₁	8	3
P ₂	9	4
P ₃	5	2
P ₄	3	1

With reference to current allocation, is system safe? If so, what is the safe sequence?

- (a) No
- (b) Yes, P₁ P₂ P₃ P₄
- (c) Yes, P₄ P₃ P₁ P₂

(d)Yes, P₂ P₁ P₃ P₄

Ans : (c) Remaining need = Max need – Current allocation.

Process	Max need	Current allocation	Remaining need
P_1	8	3	5
P_2	9	4	5
P_3	5	2	3
P_4	3	1	2
		10	

Available resources = $12 - 10 = 2$

Then, we can satisfy needs of P_4 , then release total 4, then satisfy P_3 , then release total 7, then satisfy P_1 , and P_2 similarly.

Hence, it the safe sequence: P_4, P_3, P_1, P_2 .

38. If the Disk head is located initially at track 32, find the number of disk moves required with FCFS scheduling criteria if the disk queue of I/O blocks requests are:

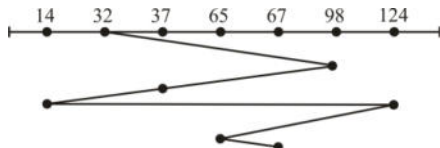
98, 37, 14, 124, 65, 67

- (a) 320 (b) 322
(c) 321 (d) 319

Ans : (c) Given I/O requests are:

98, 37, 14, 124, 65, 67

Using FCFS (first come first serve) scheduling:



Then total disk moves:

$$= (98 - 32) + (98 - 37) + (37 + 14) + (124 - 14) + (124 - 65) + (67 - 65) = 321$$

39. In UNIX, _____ creates three subdirectories: 'PIS' and two subdirectories 'progs' and 'data' from just created subdirectory 'PIS'.

- (a) mkdir PIS/progs PIS/data PIS
(b) mkdir PIS progs data
(c) mkdir PIS PIS/progs PIS/data
(d) mkdir PIS/progs data

Ans : (c) mkdir PIS/progs PIS/data.

This creates three subdirectories PIS and two subdirectories under PIS. The order of specifying the arguments is important; you obviously cannot create a subdirectory before creation of its parent directory.

40. A scheduling Algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (lowest priority). The scheduler reevaluates the process priority for every 'T' time units and decides next process to be scheduled. If the process have no I/O operations and all arrive at time

zero, then the scheduler implements _____ criteria.

- (a) Priority scheduling
- (b) Round Robin Scheduling
- (c) Shortest Job First
- (d) FCFS

Ans : (b) The scheduling algorithm works as **round robin algorithm with quantum time equals to T** . After a process's turn comes and it has executed for T units, its waiting time becomes least and its turn comes again after every other process has got the chance for T units.

41. If S_1 is total number of modules defined in the program architecture, S_3 is the number of modules whose correct function depends on prior processing then the number of modules not dependent on prior processing is:

- (a) $1 + \frac{S_3}{S_1}$
- (b) $1 - \frac{S_3}{S_1}$
- (c) $1 + \frac{S_1}{S_3}$
- (d) $1 - \frac{S_1}{S_3}$

Ans : (b) Modules not dependent on prior processing
 $D_3 = 1 - (S_3/S_1)$

42. The _____ model is preferred for software development when the requirements are not clear.

- (a) Rapid Application Development
- (b) Rational Unified Process
- (c) Evolutionary Model
- (d) Waterfall Model

Ans : (c) Evolutionary development is an iterative and incremental approach to software development. It is preferred for software development when requirements are not clear.

43. Which of the following is not included in waterfall model?

- (a) Requirement analysis
- (b) Risk analysis
- (c) Design
- (d) Coding

Ans : (b) The **waterfall model** is a sequential (non-iterative) process, it includes requirements, design, implementation, verification and maintenance.
Risk analysis used in spiral model.

44. The cyclomatic complexity of a flow graph $V(G)$, in terms of predicate node is:

- (a) $P + 1$ (b) $P - 1$
(c) $P - 2$ (d) $P + 2$

Where P is number of predicate nodes in flow graph $V(G)$.

Ans : (a) The cyclomatic complexity of a flow graph $V(G) = P + 1$, where P is number of predicate nodes.

45. The extent to which a software tolerates the unexpected problems, is termed as:

- (a) Accuracy (b) Reliability
(c) Correctness (d) Robustness

Ans : (d) Robustness is the ability of a computer system to cope with errors during execution and cope with erroneous input.

Correctness of an algorithm is asserted when it is said that the algorithm is correct with respect to a specification.

46. An attacker sits between customer and Banker, and captures the information from the customer and retransmits to the banker by altering the information. This attack is called as _____.

- (a) Masquerade Attack
(b) Replay Attack
(c) Passive Attack
(d) Denial of Service Attack

Ans : (b) (i) A replay attack (play back attack) is a form of network attack in which a valid data transmission is maliciously or fraudulently repeated or delayed.

(ii) A masquerade attack is an attack that uses a fake identity.

(iii) A passive attack is network attack in which a system is monitored and sometimes scanned for open ports and vulnerabilities.

(iv) Denial of service attack, a type of attack on a network that is designed to bring the network to its knees by flooding it with useless traffic.

47. Consider the following two statements:

(i) Business intelligence and Data warehousing is used for forecasting and Data mining.

(ii) Business intelligence and Data warehousing is used for analysis of large volumes of sales data.

Which one of the following options is correct?

- (a) (i) is true, (ii) is false.
(b) Both (i) and (ii) are true.
(c) (i) is false, (ii) is true.
(d) Both (i) and (ii) are false.

Ans: (b) (i) Business intelligence and Data warehousing is used for forecasting and Data mining.

(ii) Business intelligence and Data warehousing is used for analysis of large volumes of sales data.

Both statements are true.

48. Pipelining improves performance by:

- (a) decreasing instruction latency
- (b) eliminating data hazards
- (c) exploiting instruction level parallelism
- (d) decreasing the cache miss rate

Ans : (c) Instruction pipelining is a technique that implements a form of parallelism called instruction level parallelism with a single processor. It therefore allows faster CPU throughput that would otherwise be possible at a given clock rate.

49. Consider the following two statements:

S1: Data scrubbing is a process to upgrade the quality of data, before it is moved into Data warehouse.

S2: Data scrubbing is a process of rejecting data from data warehouse to create indexes.

Which one of the following options is correct?

- (a) S1 is true, S2 is false.
- (b) S1 is false, S2 is true.
- (c) Both S1 and S2 are false.
- (d) Both S1 and S2 are true.

Ans : (a) Data scrubbing, also called data cleaning, is the process of removing data in a database that is incorrect, incomplete, improperly formatted, or duplicated.

So, statement (i) is true and (ii) is false.

50. Given the following statements:

S1: Strategic value of data mining is time stamping.

S2: Information collection is an expensive process in building an expert system.

Which of the following options is correct?

- (a) Both S1 and S2 are false.
- (b) Both S1 and S2 are true.
- (c) S1 is true, S2 is false.
- (d) S1 is false, S2 is true.

Ans : (b)

- Strategic value of data mining is time stamping → correct
- Information collection is an expensive process in building an expert system → correct
- An expert system is a computer system that emulates the decision making ability of a human expert.